

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2– Concept Mapping (5 QUESTIONS)

1. Construct a comprehensive and well-organized concept map illustrating the complete pipeline from real-world scene capture to digital image representation. Identify all major concepts, establish meaningful relationships among intermediate transformations and dependencies, integrate insights from at least five recent literature sources, and present the map with clear hierarchy, linking phrases, and logical organization.

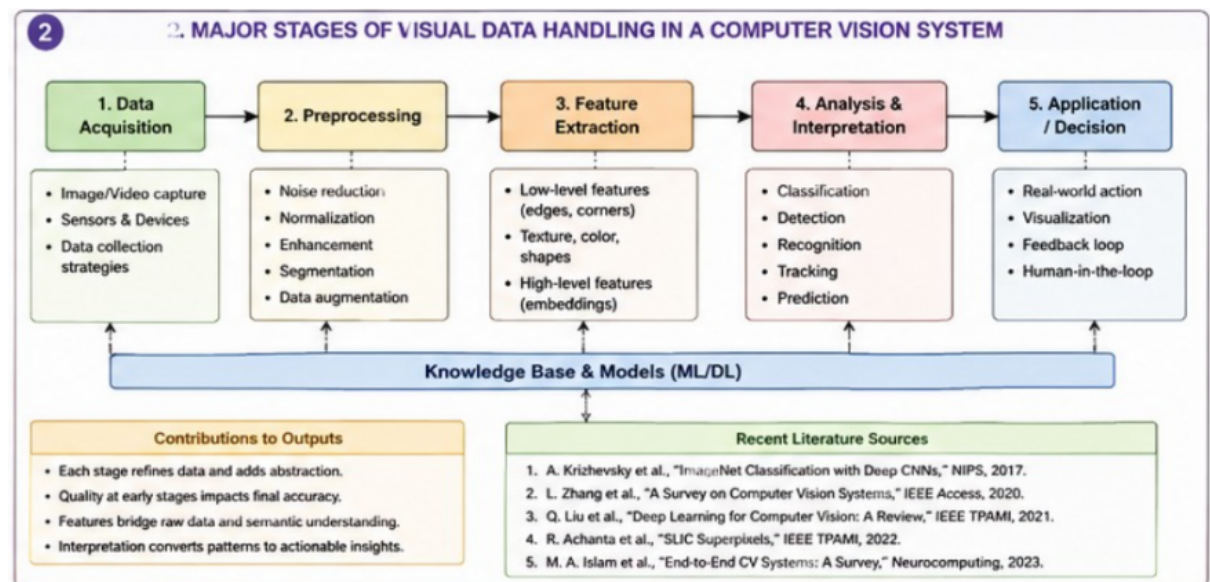
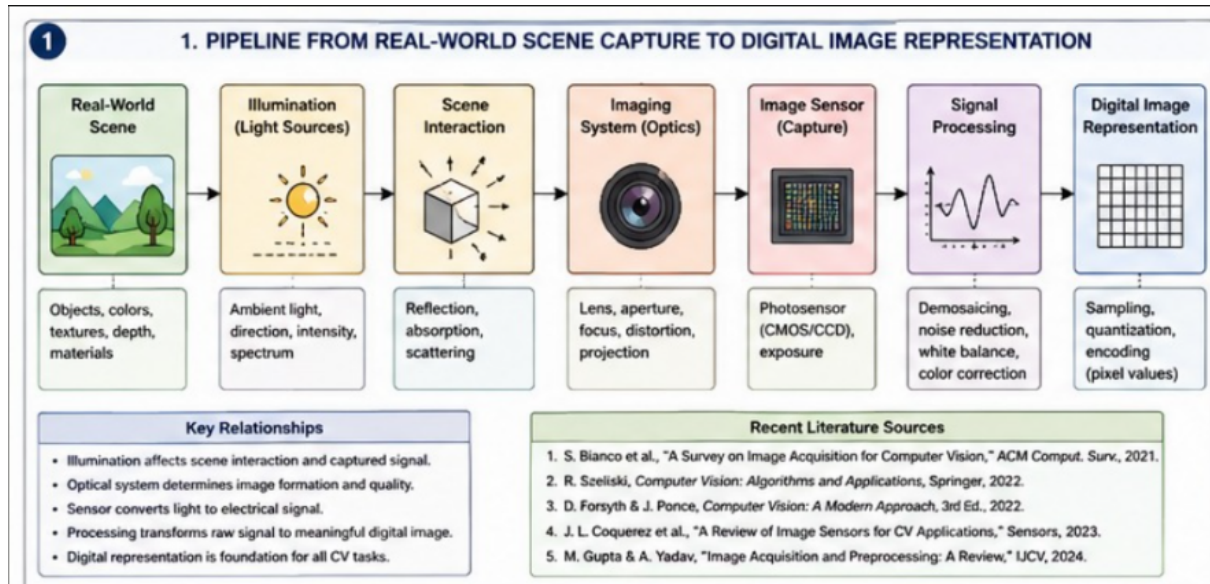
2. Develop a hierarchically structured concept map illustrating the major stages of visual data handling in a Computer Vision system. Clearly show the relationships among acquisition, preprocessing, feature extraction, analysis, and interpretation stages, explain their contributions to meaningful outputs, incorporate findings from at least five literature sources, and ensure clarity and readability.

3. Create a detailed concept map explaining the relationships among image resolution, spatial resolution, intensity resolution, and image quality. Clearly illustrate cause-and-effect relationships, discuss their impact on image usability in different applications, integrate evidence from at least five research articles, and organize the concepts logically.

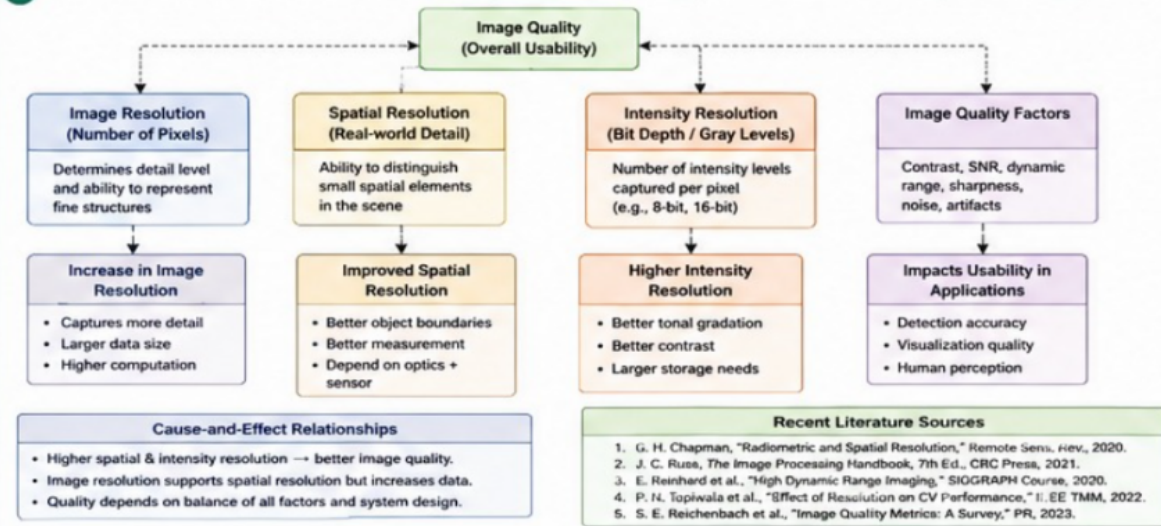
4. Design an analytically organized concept map illustrating the factors influencing image acquisition conditions, including illumination, sensor characteristics, motion, environment, and optics. Show how these factors affect downstream image processing and application performance, integrate relevant literature findings, and present the map in a visually coherent manner.

5. Construct an original concept map illustrating the relationship between Computer Vision system capabilities and the selection of appropriate vision-based applications. Identify major system capabilities, establish their influence on application effectiveness, integrate perspectives from multiple literature sources, and present the concepts using a clear hierarchical structure.

Assessment Tool -2



3. RELATIONSHIPS AMONG IMAGE RESOLUTION, SPATIAL RESOLUTION, INTENSITY RESOLUTION, AND IMAGE QUALITY



4. FACTORS INFLUENCING IMAGE ACQUISITION AND THEIR DOWNSTREAM IMPACT

